

Non-Resonant Charged-Particle Reactions: S-Factor, Gamow Peak, and Electron Screening

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July 3, 2026

Outline

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The Problem: Fusion Below the Coulomb Barrier

Charged-particle reaction cross sections fall steeply with decreasing energy because of the Coulomb repulsion between projectile and target.

- Stellar fusion occurs at energies of only a few keV to a few hundred keV;
- this is far below the Coulomb barrier V_C (typically hundreds of keV to several MeV);
- classically, such reactions could not occur at all;
- quantum mechanically, particles can **tunnel** through the classically forbidden region.

This single fact – that fusion in stars is fundamentally a tunneling phenomenon – underlies everything that follows: the Gamow factor, the S-factor, and the Gamow Peak.

Tunneling Probability and the Sommerfeld Parameter

Solving the radial Schrödinger equation for a Coulomb potential $V(r) = Z_1 Z_2 e^2 / r$ in the WKB approximation gives a barrier-penetration probability

$$P(E) \propto e^{-2\pi\eta}, \quad \eta = \frac{Z_1 Z_2 e^2}{\hbar v}.$$

η is the **Sommerfeld parameter**. Using $v = \sqrt{2E/\mu}$,

$$2\pi\eta = \frac{2\pi}{\hbar} \sqrt{\frac{\mu}{2E}} Z_1 Z_2 e^2 = 31.29 Z_1 Z_2 \sqrt{\frac{\mu}{E}} \quad (E \text{ in keV}, \mu \text{ in amu}).$$

$e^{-2\pi\eta}$ is the **Gamow factor**: the exponential suppression of low-energy charged-particle fusion due to Coulomb-barrier penetration.

Cross Section for Non-Resonant Reactions

The cross section for a non-resonant charged-particle reaction is written as the product of three physically distinct factors:

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}.$$

- ❶ $1/E$: the geometric, de Broglie-wavelength dependence of the reaction probability;
- ❷ $e^{-2\pi\eta}$: Coulomb-barrier penetration by quantum tunneling (Gamow factor);
- ❸ $S(E)$: the astrophysical S-factor, containing the nuclear-structure information.

Unlike $\sigma(E)$, which can vary by many orders of magnitude over a small energy range (e.g. $\sim 10^{-11}$ to 10^{-6} for $p+^{16}\text{O}$ across its measured range), $S(E)$ varies slowly and smoothly with energy – making it far easier to extrapolate to stellar energies.

[Ref: C. Iliadis, Wiley-VCH]

Why the S-Factor Improves Extrapolation

Laboratory measurements are rarely made at true stellar energies:

- stellar fusion typically occurs at 10–300 keV;
- laboratory measurements are usually carried out at higher energies, where cross sections are measurable;
- direct extrapolation of $\sigma(E)$ is unreliable, since small energy uncertainties translate into enormous uncertainties because of its exponential dependence.

Defining $S(E) = \sigma(E) E e^{2\pi\eta}$ removes the two dominant, rapidly varying factors (the $1/E$ geometric term and the Gamow tunneling factor), leaving a quantity dominated by nuclear-structure effects – spectroscopic properties, wavefunction overlaps, and transition matrix elements – which is far safer to extrapolate.

Reaction Rate and the Emergence of the Gamow Peak

The thermally averaged reaction rate per particle pair is

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty E \sigma(E) e^{-E/kT} dE = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E) e^{-2\pi\eta} e^{-E/kT} dE.$$

The integrand is governed by **two competing exponentials**:

- $e^{-E/kT}$ (Maxwell–Boltzmann): favours *low* energies – high-energy particles are rare;
- $e^{-2\pi\eta}$ (tunneling): favours *high* energies – barrier penetration becomes easier.

Their product has a maximum at an intermediate energy E_0 : the **Gamow Peak** – the narrow window where most stellar fusion reactions actually occur. Note $E_0 > kT$, so reactions occur in the high-energy tail of the Maxwell–Boltzmann distribution, not at its thermal peak.

Gamow Peak Centroid E_0

Setting the first derivative of the integrand $e^{-2\pi\eta - E/kT}$ with respect to E to zero gives the centroid of the Gamow Peak:

$$E_0 = 1.22 \left(Z_1^2 Z_2^2 \mu T_6^2 \right)^{1/3} \text{ keV},$$

where Z_1, Z_2 are the charges of projectile and target, μ is the reduced mass in amu, and $T_6 = T/10^6 \text{ K}$.

Reaction	E_0 (keV)
p + p	5.9
p + ^{14}N	26.5
α + ^{12}C	56.0
$^{16}\text{O} + ^{16}\text{O}$	237.0

Key trends:

- E_0 increases with $Z_1 Z_2$;
- $E_0 < V_C$ always, except $T \sim 10 \text{ GK}$;
- fusion in stars remains a tunneling phenomenon at every burning stage.

Effect of Charge on the Gamow Peak: Shift and Shrinkage

At a fixed temperature (e.g. $T = 0.03$ GK), increasing the projectile and target charges produces two simultaneous effects on the Gamow Peak:

- 1 the centroid E_0 **shifts to higher energy** ($p+p$: 5.9 keV $\rightarrow$ $p+^{12}\text{C}$: <50 keV $\rightarrow$ $^{12}\text{C}+\alpha$: ~ 0.1 MeV);
- 2 the **area under the peak decreases rapidly** – the overall reaction probability falls steeply as the Coulomb barrier grows.

Astrophysical consequence: in a stellar plasma containing a mixture of nuclei, reactions with the *smallest* Coulomb barrier contribute most of the nuclear energy generation and are consumed most rapidly – this single trend is the seed of the entire stellar burning sequence.

[Ref: C. Iliadis, Wiley-VCH]

Approximating the Gamow Peak with a Gaussian

The exact integrand is not perfectly Gaussian, but near E_0 it is very well approximated by one:

$$\exp\left(-\frac{2\pi}{\hbar} \sqrt{\frac{\mu}{2E}} Z_1 Z_2 e^2 - \frac{E}{kT}\right) \approx \exp\left(-\frac{3E_0}{kT}\right) \exp\left[-\left(\frac{E - E_0}{\Delta/2}\right)^2\right] \equiv I_{\max} \exp\left[-\left(\frac{E - E_0}{\Delta/2}\right)^2\right].$$

The same centroid and curvature are used for both curves. Separating the integrand this way isolates:

- $I_{\max} = \exp(-3E_0/kT)$ – the **peak height** (value at $E = E_0$);
- a Gaussian envelope of width Δ around E_0 .

This separation makes the reaction-rate integral analytically tractable.

Peak Height $I_{\max}$: Extreme Sensitivity to Charge

$$I_{\max} = \exp\left(-\frac{3E_0}{kT}\right)$$

falls dramatically as E_0 (and hence the Coulomb barrier) increases:

At $T = 15$ MK:

Reaction	$I_{\max}$
p + p	10^{-6}
p + ^{14}N	10^{-27}
α + ^{12}C	10^{-57}
$^{16}\text{O} + ^{16}\text{O}$	10^{-239}

Even a modest increase in Coulomb barrier reduces reaction probability by *many* orders of magnitude – this is why stellar fusion rates are extraordinarily sensitive to both nuclear charge and temperature.

Peak Width Δ and the Gamow Window

Matching second derivatives of the exact integrand and its Gaussian approximation at E_0 gives the **peak width**:

$$\Delta = \frac{4}{\sqrt{3}} \sqrt{E_0 kT} = 0.749 \left(Z_1^2 Z_2^2 \mu T_6^5 \right)^{1/6} \text{ keV}.$$

Key points:

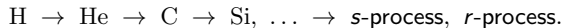
- $\Delta < E_0$ always, since $kT \ll E_0$ for charged-particle reactions;
- Δ *increases* with Coulomb barrier, even as the peak *area* decreases;
- most reactions occur within the **Gamow Window**, $E_0 - \Delta/2 \leq E \leq E_0 + \Delta/2$;
- with increasing $Z_1 Z_2$ (e.g. $p+p \rightarrow \text{Li} \rightarrow \text{C} \rightarrow \text{Ne} \rightarrow \text{Ca} \rightarrow \text{Ru}$), this window shifts to higher energy *and* becomes broader.

The Gamow window is often too low for direct laboratory measurement – this is precisely why S-factor extrapolation is essential.

Burning Sequence: Consequence of Falling Peak Area

At any instant, a stellar plasma contains a mixture of nuclear species. Reactions proceed in a strict, well-defined sequence governed by Coulomb barrier height:

- 1 nuclei with the **smallest** Coulomb barrier are consumed first, dominating energy production at that stage;
- 2 once exhausted, gravitational contraction raises the core temperature until the next-lowest-barrier fuel ignites;
- 3 the energy released stabilizes the star against further contraction;
- 4 the cycle repeats for progressively higher-charge fuels.



This contract–heat–ignite cycle, paced entirely by rising Coulomb barriers, defines the well-separated burning epochs of stellar evolution.

Closed-Form Reaction Rate

Replacing the Gamow Peak with its Gaussian approximation and treating $S(E) \approx S(E_0) \equiv S_0$ as constant (extending the lower limit to $-\infty$ at negligible cost):

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi \mu}} \frac{1}{(kT)^{3/2}} S_0 \int_0^\infty e^{-2\pi\eta} e^{-E/kT} dE = \sqrt{\frac{2}{\mu}} \frac{\Delta}{(kT)^{3/2}} S_0 e^{-3E_0/kT}.$$

Substituting the expressions for E_0 and Δ gives the standard numerical form:

$$\langle \sigma v \rangle = 7.2 \times 10^{-19} \frac{S_0}{\mu Z_1 Z_2} \tau^2 e^{-\tau} \text{ cm}^3 \text{ s}^{-1}, \quad \tau = \frac{3E_0}{kT},$$

with S_0 in keV·barn and μ in amu. **Two simplifying assumptions** were made: a perfectly Gaussian peak, and an energy-independent S-factor – both require small corrections in precision work.

Extreme Sensitivity to Temperature: The Stellar Thermostat

Since $\langle\sigma v\rangle \propto \tau^2 e^{-\tau}$, a power-law fit locally gives $\langle\sigma v\rangle \propto T^n$:

	Reaction	$\langle\sigma v\rangle$	V_C (MeV)
At $T = 15$ MK:	$p+p$	$\propto T^{3.9}$	0.55
	$p+^{14}\text{N}$	$\propto T^{20}$	2.27
	$\alpha+^{12}\text{C}$	$\propto T^{42}$	3.43
	$^{16}\text{O}+^{16}\text{O}$	$\propto T^{182}$	14.07

This thermostat mechanism is why stars remain stable over millions to billions of years despite the extreme sensitivity of $\langle\sigma v\rangle$ to T and V_C .

Negative-feedback stabilization:

- a small T rise causes a huge rise in fusion rate and energy output;
- the excess energy drives expansion;
- expansion cools the core, reducing the rate again.

Electron Screening: Bare vs. Shielded Nucleus

Inside a star, bare nuclei react with each other (no surrounding electrons). In a laboratory target, however, nuclei exist as neutral atoms, surrounded by an electron cloud that partially screens the nuclear charge.

- total potential $V_{\text{tot}} = Z_1 e/r - Z_1 e/R_a$ goes to zero outside the atomic radius R_a , rather than asymptoting slowly as for a bare nucleus;
- the projectile therefore experiences a **reduced effective Coulomb barrier**:

$$E_{\text{eff}} = \frac{Z_1 Z_2 e^2}{R_n} - \frac{Z_1 Z_2 e^2}{R_a}, \quad \frac{R_n}{R_a} \sim 10^{-5};$$

- barrier penetration becomes easier, slightly enhancing the measured cross section.

Quantifying the Screening Correction

Penetration through a shielded Coulomb potential at projectile energy E_s is equivalent to penetration through a bare nucleus at energy $E = E_s + U_e$, with screening energy

$$U_e = \frac{Z_1 Z_2 e^2}{R_a}.$$

The enhancement ratio in cross section is $m = \sigma(E)/\sigma(E_s)$. For p+p:

E_s (keV)	enhancement m
1	33%
5	2.3%
20	0.21%
100	0.0027%

Since the Gamow window lies at higher energies than where screening matters, shielding corrections can be discarded – laboratory measurements can be treated as if for bare nuclei.

Electron Screening in Dense Stellar Plasmas

Although negligible in typical laboratory conditions, screening becomes important at very high stellar densities, where many electrons cluster around each nucleus.

- a denser electron cloud further reduces the effective Coulomb barrier;
- this eases penetration and **increases** the cross section;
- the screened and bare reaction rates are related by

$$\langle \sigma v \rangle_{\text{shielded}} = f \langle \sigma v \rangle_{\text{bare}}, \quad 1 < f < 2,$$

with f increasing at higher plasma density.

Electron screening therefore generally *enhances* stellar reaction rates, with importance growing alongside plasma density.

Summary (1/2): S-Factor and Gamow Peak

- Charged-particle fusion in stars proceeds via quantum tunneling: $P(E) \propto e^{-2\pi\eta}$, $\eta = Z_1 Z_2 e^2 / \hbar v$.
- $\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}$ separates geometric, tunneling, and nuclear-structure effects; the smoothly varying $S(E)$ is the safer quantity to extrapolate to stellar energies.
- The competition between $e^{-E/kT}$ and $e^{-2\pi\eta}$ produces the **Gamow Peak**, centred at $E_0 = 1.22(Z_1^2 Z_2^2 \mu T_6^2)^{1/3}$ keV.
- Increasing $Z_1 Z_2$ shifts E_0 to higher energy and shrinks the peak area, governing the stellar burning sequence.

Summary (2/2): Width, Rate, and Screening

- Approximating the peak as Gaussian gives height $I_{\max} = e^{-3E_0/kT}$ and width $\Delta = \frac{4}{\sqrt{3}}\sqrt{E_0 kT}$, with $\Delta < E_0$ always.
- Most reactions occur in the **Gamow Window**, $E_0 \pm \Delta/2$ – often below direct measurement, motivating S-factor extrapolation.
- Closed-form reaction rate: $\langle\sigma v\rangle = 7.2 \times 10^{-19} \frac{S_0}{\mu Z_1 Z_2} \tau^2 e^{-\tau}$, $\tau = 3E_0/kT$ – extremely sensitive to T and V_C , providing the stellar thermostat.
- Electron screening lowers the effective barrier; negligible in the lab (Gamow window lies above U_e) but can enhance rates by factor f ($1 < f < 2$) in dense stellar plasmas.